

Class 12: RNASeq with DESeq2

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Background

Today we will perform a RNASeq analysis of the effects of a common steroid on airway cells.

In particular, we will be looking at dexamethasone (hereafter just called “dex”) on different airway smooth muscle cells lines (ASM cells).

Data Import

We need two different inputs:

- **countData**: with genes in rows and experiments in columns
- **callData**: meta data that describes the columns in countData.

```
counts <- read.csv("https://bioboot.github.io/bimm143_W26/class-material/airway_scaledcounts.csv")
calls <- read.csv("https://bioboot.github.io/bimm143_W26/class-material/airway_metadata.csv")
```

A look at counts and metadata.

```
head(counts)
```

	SRR1039508	SRR1039509	SRR1039512	SRR1039513	SRR1039516
ENSG000000000003	723	486	904	445	1170
ENSG000000000005	0	0	0	0	0
ENSG000000000419	467	523	616	371	582
ENSG000000000457	347	258	364	237	318
ENSG000000000460	96	81	73	66	118
ENSG000000000938	0	0	1	0	2

	SRR1039517	SRR1039520	SRR1039521
ENSG000000000003	1097	806	604
ENSG000000000005	0	0	0
ENSG000000000419	781	417	509
ENSG000000000457	447	330	324
ENSG000000000460	94	102	74
ENSG000000000938	0	0	0

```
head(calls)
```

	id	dex	celltype	geo_id
1	SRR1039508	control	N61311	GSM1275862
2	SRR1039509	treated	N61311	GSM1275863
3	SRR1039512	control	N052611	GSM1275866
4	SRR1039513	treated	N052611	GSM1275867
5	SRR1039516	control	N080611	GSM1275870
6	SRR1039517	treated	N080611	GSM1275871

Q1: How many genes are in this dataset?

```
nrow(counts)
```

```
[1] 38694
```

Q2: How many 'control' cell lines do we have?

```
sum (calls$dex == "control")
```

```
[1] 4
```

Differential Gene Expression

We have four replicate drug treated and control (no drug) columns/experiments in our `countdata` object.

We want one mean value for each gene (rows) in “treated” and one mean value for each in “control” columns.

Step 1. Find all “control” columns.

```
control.inds <- calls$dex == "control"
```

Step 2. Extract these columns from the call data to a new object called `controlledcounts`

```
controlledcounts <- counts[,control.inds]
```

Step 3. Then calculate the mean value for each gene.

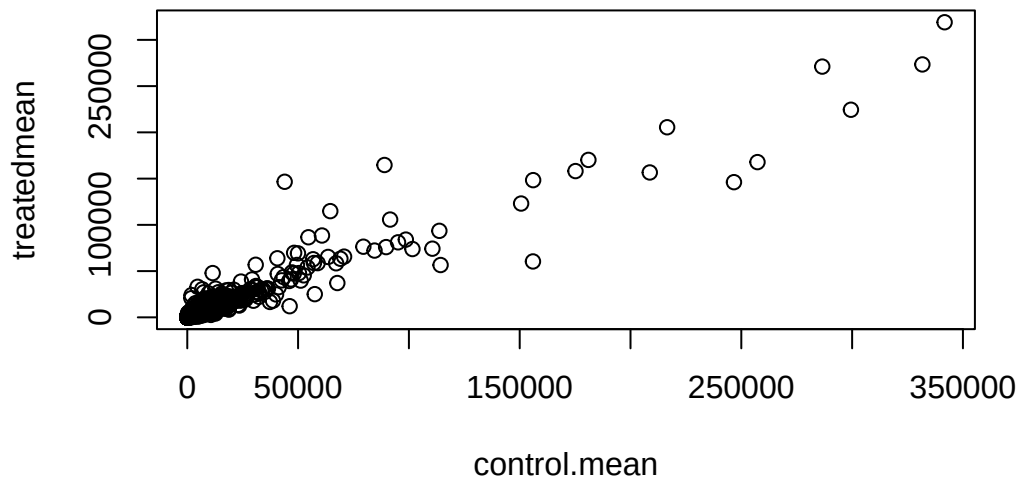
```
control.mean <- rowMeans(controlledcounts)
```

Now do the same thing for the treated experiments.

```
treatedinds <- calls$dex == "treated"  
treatedcounts <- counts[,treatedinds]  
treatedmean <- rowMeans(treatedcounts)
```

Put these together for easy book-keeping as `meancounts`, and a quick plot.

```
meancounts <- data.frame(control.mean, treatedmean)  
plot(meancounts)
```

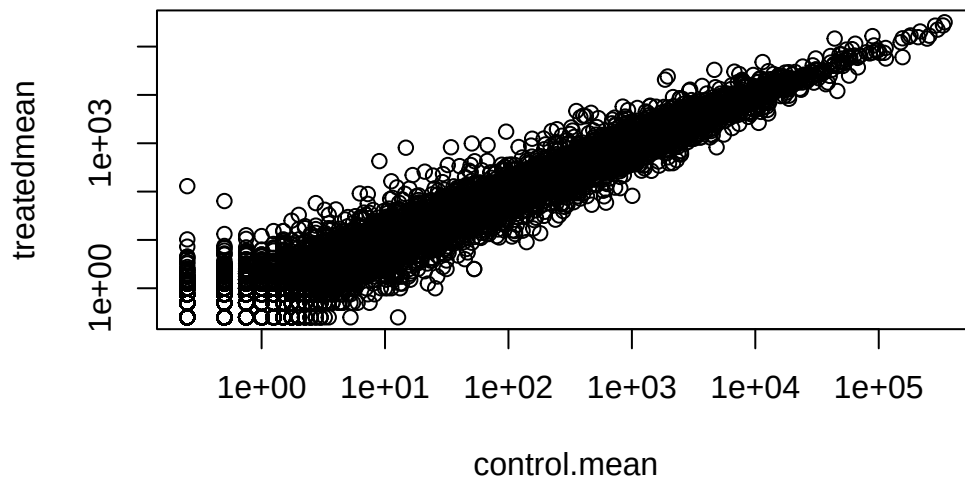


Let's log transform this count data:

```
plot(meancounts, log="xy")
```

Warning in xy.coords(x, y, xlabel, ylabel, log): 15032 x values <= 0 omitted from logarithmic plot

Warning in xy.coords(x, y, xlabel, ylabel, log): 15281 y values <= 0 omitted from logarithmic plot



N.B. We most often use $\log_2$ for this type of data as it makes the interpretation much more straightforward. Treated as a “fold-change”.

If there was no change we would have a $\log_2$ -fc of zero:

```
log2(10/10)
```

```
[1] 0
```

If we had double the amount of transcript around:

```
log2(20/10)
```

```
[1] 1
```

If we had half as much transcript around we would have a $\log_2$ -fc of -1

```
log2(5/10)
```

```
[1] -1
```

Q. Calculate a log2 fold change value for all our genes and add it as a new column to our `meancounts` object.

```
meancounts$log2fc <- log2(meancounts$treatedmean/meancounts$control.mean)
head(meancounts)
```

	control.mean	treatedmean	log2fc
ENSG000000000003	900.75	658.00	-0.45303916
ENSG000000000005	0.00	0.00	NaN
ENSG000000000419	520.50	546.00	0.06900279
ENSG000000000457	339.75	316.50	-0.10226805
ENSG000000000460	97.25	78.75	-0.30441833
ENSG000000000938	0.75	0.00	-Inf

There are some funky log2fc values (NaN and -Inf) here that come about when ever we have 0 mean count values. Typically we would remove these genes from any further analysis - as we can't say anything about them if we have no data for them.

##DESeq Analysis

Let's do this analysis with an estimate of statistical significance using **DESeq2** package.

```
library(DESeq2)
```

Loading required package: S4Vectors

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics

Attaching package: 'generics'

The following objects are masked from 'package:base':

```
as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
setequal, union
```

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

anyDuplicated, aperm, append, as.data.frame, basename, cbind,
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
unsplit, which.max, which.min

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

expand.grid, I, unname

Loading required package: IRanges

Attaching package: 'IRanges'

The following object is masked from 'package:grDevices':

windows

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

colAlls, colAnyNAs, colAnys, colAvgPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Welcome to Bioconductor

Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

rowMedians

The following objects are masked from 'package:matrixStats':

anyMissing, rowMedians

DESeq (like many bioconductor packages) wants its input data in a very specific data.

```
dds <- DESeqDataSetFromMatrix(countData = counts,  
                              colData = calls,  
                              design= ~dex)
```

converting counts to integer mode

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

Run the DESeq analysis pipeline:

The main function DESeq()

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
res <- results(dds)  
head(res)
```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

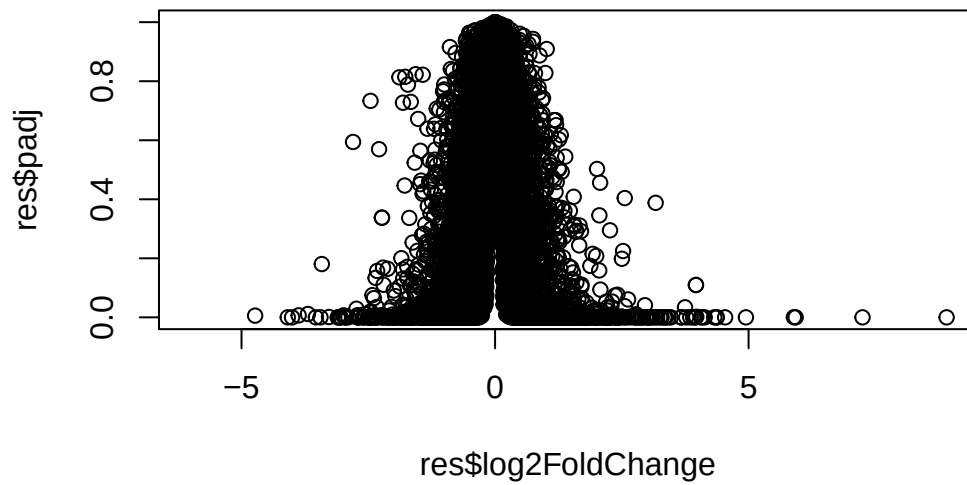
DataFrame with 6 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG000000000003	747.194195	-0.3507030	0.168246	-2.084470	0.0371175
ENSG000000000005	0.000000	NA	NA	NA	NA
ENSG000000000419	520.134160	0.2061078	0.101059	2.039475	0.0414026
ENSG000000000457	322.664844	0.0245269	0.145145	0.168982	0.8658106
ENSG000000000460	87.682625	-0.1471420	0.257007	-0.572521	0.5669691
ENSG000000000938	0.319167	-1.7322890	3.493601	-0.495846	0.6200029
	padj				
	<numeric>				
ENSG000000000003	0.163035				
ENSG000000000005	NA				
ENSG000000000419	0.176032				
ENSG000000000457	0.961694				
ENSG000000000460	0.815849				
ENSG000000000938	NA				

Volcano Plot

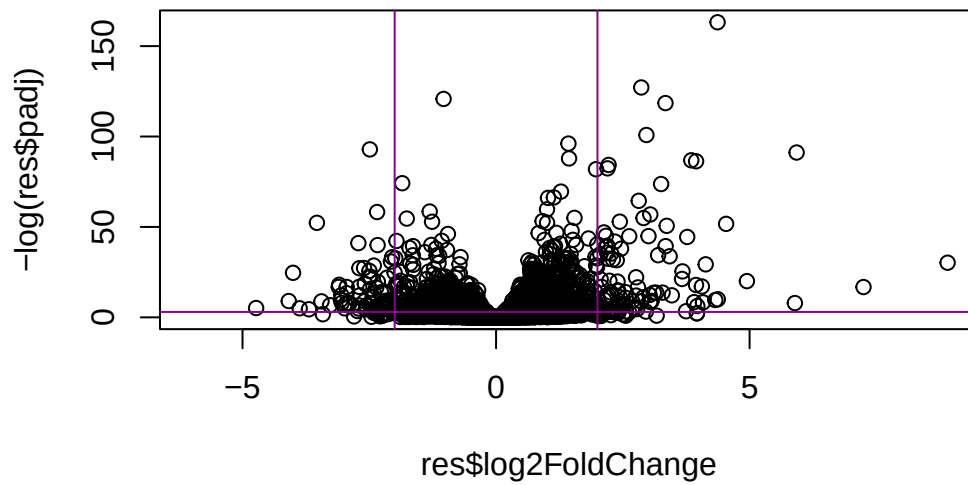
This is a main summary results figure from these kinds of studies. It is a plot of log2 fold-change vs. P-value.

```
plot(res$log2FoldChange,  
     res$padj)
```



Again this y-axis is highly skewed and need log transformation, we can flip the y-axis with a minus sign so it looks like every other volcano plot.

```
plot(res$log2FoldChange,  
      -log(res$padj))  
abline(v=-2, col="magenta4")  
abline(v=+2, col="magenta4")  
abline(h=-log(0.05), col="magenta4")
```

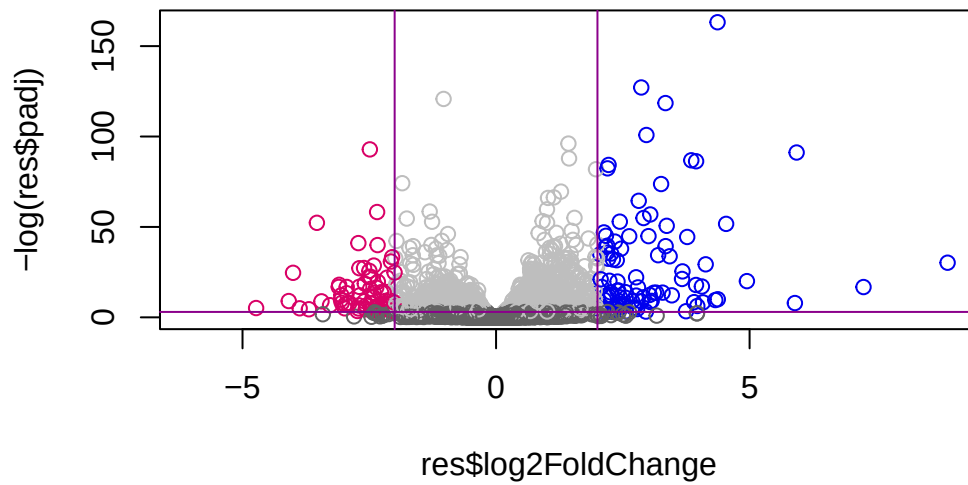


Adding some color annotation to this plot.

Start with a default base.

```
mycols <- rep("grey", nrow(res))
mycols[res$log2FoldChange > 2] <- "blue2"
mycols[res$log2FoldChange < -2] <- "#D90166"
mycols[res$padj >= 0.05 ] <- "#636363"

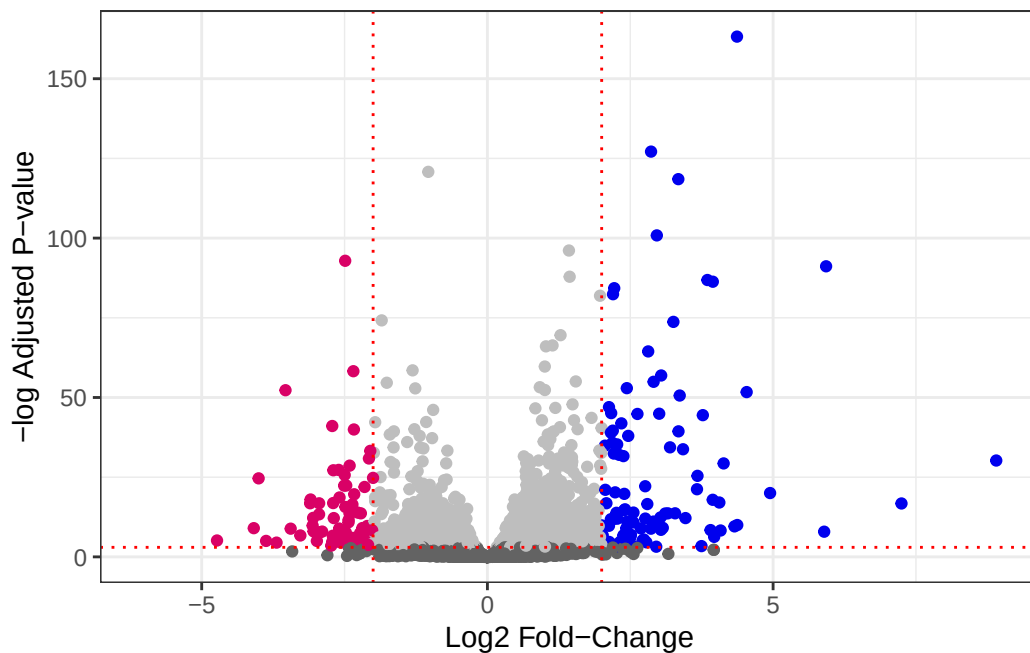
plot(res$log2FoldChange,
      -log(res$padj),
      col=mycols)
abline(v=-2, col="magenta4")
abline(v=+2, col="magenta4")
abline(h=-log(0.05), col="magenta4")
```



Q. Make a presentation quality ggplot version of this plot. Include clear axis labels, a clean theme, your custom colors, cut-off lines, and a plot title.

```
library(ggplot2)
ggplot(res) +
  aes(log2FoldChange, -log(padj)) +
  geom_point(
    colour = mycols) +
  labs(x="Log2 Fold-Change",
       y="-log Adjusted P-value") +
  theme_bw() +
  geom_vline(xintercept= c(2,-2),
             colour="red",
             linetype = "dotted") +
  geom_hline(yintercept = -log(0.05),
             colour="red",
             linetype="dotted")
```

Warning: Removed 23549 rows containing missing values or values outside the scale range (`geom\_point()`).



Save our results.

Write a CSV file.

```
write.csv(res, file="results.csv")
```

Add annotation data

We need to add missing annotation to our main `res` results object. This includes the common gene “symbol”.

```
head(res)
```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

DataFrame with 6 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG000000000003	747.194195	-0.3507030	0.168246	-2.084470	0.0371175
ENSG000000000005	0.000000	NA	NA	NA	NA
ENSG000000000419	520.134160	0.2061078	0.101059	2.039475	0.0414026

ENSG00000000457	322.664844	0.0245269	0.145145	0.168982	0.8658106
ENSG00000000460	87.682625	-0.1471420	0.257007	-0.572521	0.5669691
ENSG00000000938	0.319167	-1.7322890	3.493601	-0.495846	0.6200029
	padj				
	<numeric>				
ENSG00000000003	0.163035				
ENSG00000000005	NA				
ENSG00000000419	0.176032				
ENSG00000000457	0.961694				
ENSG00000000460	0.815849				
ENSG00000000938	NA				

We will use R and bioconductor to do this “ID mapping”.

```
library("AnnotationDbi")
library("org.Hs.eg.db")
```

Let’s see what databases we can use for translation/mapping...

```
columns(org.Hs.eg.db)
```

[1]	"ACCNUM"	"ALIAS"	"ENSEMBL"	"ENSEMBLPROT"	"ENSEMBLTRANS"
[6]	"ENTREZID"	"ENZYME"	"EVIDENCE"	"EVIDENCEALL"	"GENENAME"
[11]	"GENETYPE"	"GO"	"GOALL"	"IPI"	"MAP"
[16]	"OMIM"	"ONTOLOGY"	"ONTOLOGYALL"	"PATH"	"PFAM"
[21]	"PMID"	"PROSITE"	"REFSEQ"	"SYMBOL"	"UCSCKG"
[26]	"UNIPROT"				

We can use the `mapIds()` function now to “translate” between any of these databases.

```
res$symbol <- mapIds(org.Hs.eg.db,
                     keys=row.names(res), # Our genenames
                     keytype="ENSEMBL",   # The format of our genenames
                     column="SYMBOL",     # The new format we want to add
                     multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
head(res)
```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

DataFrame with 6 rows and 7 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG000000000003	747.194195	-0.3507030	0.168246	-2.084470	0.0371175
ENSG000000000005	0.000000	NA	NA	NA	NA
ENSG000000000419	520.134160	0.2061078	0.101059	2.039475	0.0414026
ENSG000000000457	322.664844	0.0245269	0.145145	0.168982	0.8658106
ENSG000000000460	87.682625	-0.1471420	0.257007	-0.572521	0.5669691
ENSG000000000938	0.319167	-1.7322890	3.493601	-0.495846	0.6200029
	padj	symbol			
	<numeric>	<character>			
ENSG000000000003	0.163035	TSPAN6			
ENSG000000000005	NA	TNMD			
ENSG000000000419	0.176032	DPM1			
ENSG000000000457	0.961694	SCYL3			
ENSG000000000460	0.815849	FIRRM			
ENSG000000000938	NA	FGR			

Q11. Also add “ENTREZID”, “GENENAME”.

```
res$entrez <- mapIds(org.Hs.eg.db,  
  keys=row.names(res),  
  column="ENTREZID",  
  keytype="ENSEMBL",  
  multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$genename <- mapIds(org.Hs.eg.db,  
  keys=row.names(res),  
  column="GENENAME",  
  keytype="ENSEMBL",  
  multiVals="first")
```

'select()' returned 1:many mapping between keys and columns


```
head(res)
```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

DataFrame with 6 rows and 9 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG000000000003	747.194195	-0.3507030	0.168246	-2.084470	0.0371175
ENSG000000000005	0.000000	NA	NA	NA	NA
ENSG000000000419	520.134160	0.2061078	0.101059	2.039475	0.0414026
ENSG000000000457	322.664844	0.0245269	0.145145	0.168982	0.8658106
ENSG000000000460	87.682625	-0.1471420	0.257007	-0.572521	0.5669691
ENSG000000000938	0.319167	-1.7322890	3.493601	-0.495846	0.6200029
	padj	symbol	entrez	genename	
	<numeric>	<character>	<character>	<character>	
ENSG000000000003	0.163035	TSPAN6	7105	tetraspanin 6	
ENSG000000000005	NA	TNMD	64102	tenomodulin	
ENSG000000000419	0.176032	DPM1	8813	dolichyl-phosphate m..	
ENSG000000000457	0.961694	SCYL3	57147	SCY1 like pseudokina..	
ENSG000000000460	0.815849	FIRRM	55732	FIGNL1 interacting r..	
ENSG000000000938	NA	FGR	2268	FGR proto-oncogene, ..	

Save annotated results to a CSV file

```
write.csv(res, file="resultsannotated.csv")
```

Pathway Analysis

What known biological pathways do our differentially expressed genes overlap with (i.e. play a role in)?

There are lots of bioconductor packages to do this type of analysis. We will use one of the oldest called **gage** along with **pathview** to render nice pictures of the pathways we find.

We can install these with the command `> BiocManager::install( c("pathview", "gage", "gageData") )`.

```
library(pathview)
```

```
#####
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
formally cite the original Pathview paper (not just mention it) in publications
or products. For details, do citation("pathview") within R.
```

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG license agreement (details at <http://www.kegg.jp/kegg/legal.html>).

```
#####
```

```
library(gage)
```

```
library(gageData)
```

```
data(kegg.sets.hs)
```

```
# Examine the first 2 pathways in this kegg set for humans
```

```
head(kegg.sets.hs, 2)
```

```
$`hsa00232 Caffeine metabolism`
```

```
[1] "10" "1544" "1548" "1549" "1553" "7498" "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
```

```
[1] "10" "1066" "10720" "10941" "151531" "1548" "1549" "1551"
[9] "1553" "1576" "1577" "1806" "1807" "1890" "221223" "2990"
[17] "3251" "3614" "3615" "3704" "51733" "54490" "54575" "54576"
[25] "54577" "54578" "54579" "54600" "54657" "54658" "54659" "54963"
[33] "574537" "64816" "7083" "7084" "7172" "7363" "7364" "7365"
[41] "7366" "7367" "7371" "7372" "7378" "7498" "79799" "83549"
[49] "8824" "8833" "9" "978"
```

The main `gage()` function that does the actual work wants a single vector as input. The KEGG data base uses ENTREZ ids so we need to provide these in ur input vector for `gage`.

```
foldchanges <- res$log2FoldChange
names(foldchanges) <- res$entrez
```

Now we can run `gage()`.

```
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

What is in the output object keggres

```
attributes(keggres)
```

```
$names
[1] "greater" "less"    "stats"
```

```
head(keggres$less)
```

	p.geomean	stat.mean
hsa05332 Graft-versus-host disease	0.0004250461	-3.473346
hsa04940 Type I diabetes mellitus	0.0017820293	-3.002352
hsa05310 Asthma	0.0020045888	-3.009050
hsa04672 Intestinal immune network for IgA production	0.0060434515	-2.560547
hsa05330 Allograft rejection	0.0073678825	-2.501419
hsa04340 Hedgehog signaling pathway	0.0133239547	-2.248547
	p.val	q.val
hsa05332 Graft-versus-host disease	0.0004250461	0.09053483
hsa04940 Type I diabetes mellitus	0.0017820293	0.14232581
hsa05310 Asthma	0.0020045888	0.14232581
hsa04672 Intestinal immune network for IgA production	0.0060434515	0.31387180
hsa05330 Allograft rejection	0.0073678825	0.31387180
hsa04340 Hedgehog signaling pathway	0.0133239547	0.47300039
	set.size	expl
hsa05332 Graft-versus-host disease	40	0.0004250461
hsa04940 Type I diabetes mellitus	42	0.0017820293
hsa05310 Asthma	29	0.0020045888
hsa04672 Intestinal immune network for IgA production	47	0.0060434515
hsa05330 Allograft rejection	36	0.0073678825
hsa04340 Hedgehog signaling pathway	56	0.0133239547

We can use **pathview** package to render a figure of any of these pathways along with annotation for our DEGs.

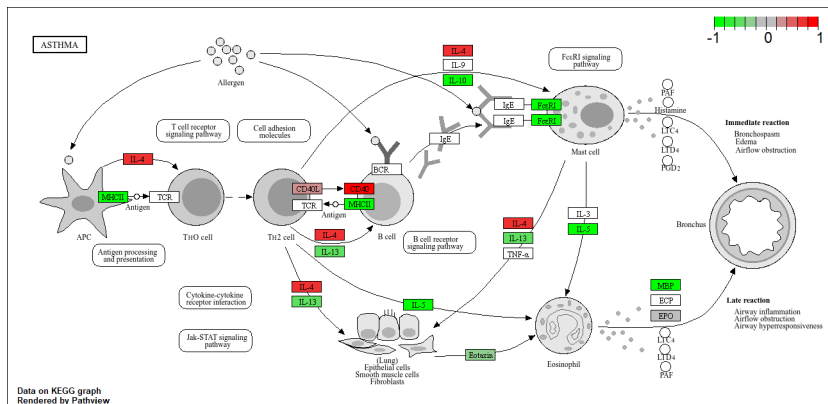
Let's see the hsa05310 Asthma pathway with our DEGs colored up.

```
pathview(gene.data = foldchanges, pathway.id = "hsa05310")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/Diego/OneDrive/Desktop/BIMM143/Class 13

Info: Writing image file hsa05310.pathview.png



Q. Can you render and insert the pathway for Graft-vs-Host disease and type 1 diabetes?

```
pathview(gene.data = foldchanges, pathway.id = c("hsa05332", "hsa04940"))
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/Diego/OneDrive/Desktop/BIMM143/Class 13

Info: Writing image file hsa05332.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/Diego/OneDrive/Desktop/BIMM143/Class 13

Info: Writing image file hsa04940.pathview.png

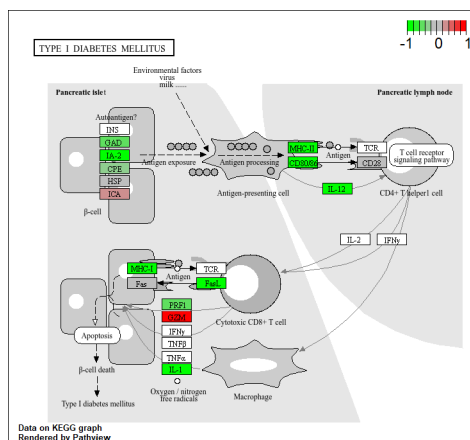


Figure 1: Pathway for Type I Diabetes

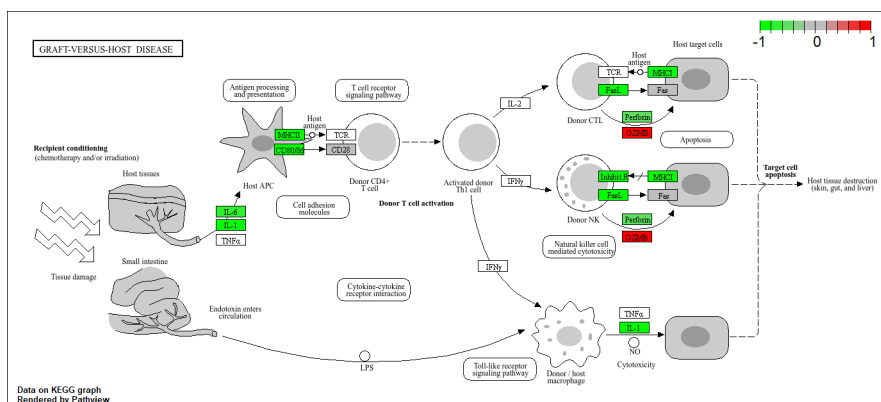


Figure 2: Pathway for Graft-versus-Host Disease